NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

Introduction to Internet of Things

Assignment-Week 1

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **15** **Total marks: 15 X 1= 15**

QUESTION 1:

What is the full form of IoT?

a. Internet of Tasks
b. Internet of Articles
c. Internet of Taps
d. None of these

Correct Answer: d. None of these

Detailed Solution: The full form of IoT is "Internet of Things"

See lecture 1 (Introduction to IoT – Part - I) @ 1:30

QUESTION 2:

The "Source Address" filed in the IPv6 header is of _____ bits in length.

a. 32

b. 64

c. 128

d. None of these

Correct Answer: c. 128

Detailed Solution: The IPv6 header uses 128 bits to represent the “Source Address”.

See lecture 2 (Introduction to IoT – Part - II) @ 18:03

QUESTION 3:

Which of the following is/are not an enabler of IoT?

a. RFID

b. Sensors and Actuators

c. Cloud Computing

d. None of these

Correct Answer: d. None of these

Detailed Solution: *See Lecture 1 @ 12:46*

QUESTION 4:

State true or false.

The gateway is tasked with assigning a unique address to each of the field devices or sensors within its jurisdiction.

a. True

b. False

Correct Answer: a. True

Detailed Solution: *See Lecture 5 @ 12:11*

QUESTION 5:

State whether the following statement is True or False.

Statement: The increasing number of devices in IoT is expected to result in an address crunch.

a. True

b. False

Correct Answer: a. True

Detailed Solution: The increasing number of devices in IoT is expected to result in an address crunch.

See lecture 2 (Introduction to IoT – Part - II) @ 01:19

QUESTION 6:

In _____ a node/network is connected to multiple networks for improved reliability.

a. Transparent roaming
b. Multi-coulding
c. None of these
d. Both (a) and (b)

Correct Answer: c. None of these

Detailed Solution: In multi-homing, a node/network is connected to multiple networks for improved reliability.

See lecture 2 (Introduction to IoT – Part - II) @ 15:22

QUESTION 7:

Which of the following is NOT a function of an IoT Gateway?

a. Connects the IoT LAN to a WAN

b. Can implement several LAN and WAN

c. Both (a) and (b)

d. Neither (a) Nor (b)

Correct Answer: d. Neither (a) Nor (b)

Detailed Solution: The functionalities of an IoT Gateway are –

a. Connects the IoT LAN to a WAN

b. Can implement several LAN and WAN

c. Forwards packets between LAN and WAN on the IP layer

See lecture 2 @ 3:09

QUESTION 8:

Which of the following statements is/are incorrect?

Statement I: An actuator is a component of a machine or system that moves or controls the mechanism or the system.

Statement II: An actuator requires a control signal and a source of energy.

a. Statement I is incorrect
b. Statement II is incorrect
c. Both statements are incorrect
d. Both statements are correct

Correct Answer: d. Both statements are correct

Detailed Solution: See lecture 04 @ 04:00

QUESTION 9:

Which of the following is NOT a function of an IoT LAN?

a. Long range communication, global
b. World wide connections
c. Both (a) and (b)
d. Neither (a) Nor (b)

Correct Answer: c. Both (a) and (b)

Detailed Solution: The functionalities of an IoT Gateway are –

a. Local, short-range communication
b. Spreads across buildings or organization

See lecture 2 (Introduction to IoT – Part - II) @ 3:09

QUESTION 10:

State whether the following statement is True or False.

Statement: The word "Transducer" is a collective term used for both Sensors and Actuators.

a. True
b. False

Correct Answer: a. True

Detailed Solution: The word "Transducer" is the collective term used for both Sensors, which can be used to sense a wide range of different energy forms such as movement, electrical signals, radiant energy, thermal, or magnetic energy, etc., and Actuators, which can be used to switch voltages or currents.

See lecture 3 (Sensing) @ 11:39

QUESTION 11:

State True or False.

Analog Sensors produce a discontinuous output signal or voltage, which is generally proportional to the quantity being measured.

a. True
b. False

Correct Answer: b. False

Detailed Solution: Analog Sensors produce a discontinuous output signal or voltage, which is generally proportional to the quantity being measured.

See lecture 3 (Sensing) @ 15:24

QUESTION 12:

If the output signal differs from the correct value by a constant, the sensor has an _____.

a. Offset error
b. Bias
c. Both (a) and (b)
d. None of these

Correct Answer: c. Both (a) and (b)
Detailed Solution: If the output signal differs from the correct value by a constant, the sensor has an offset error or bias.

See lecture 3 (Sensing) @ 20:33

QUESTION 13:

State True or False

Pneumatic actuators enable the production of large forces from a relatively larger pressure changes.

a. False
b. True

Correct Answer: a. False

Detailed Solution: Pneumatic actuators enable the production of large forces from a relatively larger pressure changes.

See lecture 4 (Actuation) @ 09:04

QUESTION 14:

A sensor node is made up of which of the following?

a. Sensor/Sensors
b. A processing unit
c. A power unit
d. All of these

Correct Answer: d. All of these

Detailed Solution: A sensor node is made up of a combination of sensor/sensors, a processor unit, a radio unit, and a power unit.

See Page number – 101, Chapter - 5, Book - Introduction to IoT, Authors – Sudip Misra, Anandarup Mukherjee, and Arijit Roy, Publisher – Cambridge University Press, Edition – 1 (2021)

QUESTION 15:

What is the full form of SMP in the context of IoT actuators?

a. Shape Memory Polymers
b. Shape Master Polymers
c. Switch Mode Power
d. Sensor Making Polymer

Correct Answer: a. Shape Memory Polymers

Detailed Solution: The full form of SMP is Shape Memory Polymers.

See Page number – 111, Chapter - 5, Book - Introduction to IoT, Authors – Sudip Misra, Anandarup Mukherjee, and Arijit Roy, Publisher – Cambridge University Press, Edition – 1 (2021)

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